Griffin Kriebel

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Education:

University of California, Davis — *Bachelor of Science, Biological Systems Engineering* **Expected Dec 2026**

ABET accredited coursework in Mechanical, Environmental, and Biotechnical Engineering

Experience:

Agnetix LED Liquid Cooling Heat Recycler, Davis, CA — *Student Researcher* **Sep 2025 - June 2026**

- UC Davis senior design project partnered with Agnetix using liquid cooling and heat transfer of LEDs to increase energy efficiency and crop productivity for indoor vertical farms.
- PLC automated temperature control where root solution is kept between 25-30℃ using recycled LED heat.
- Excess heat is removed from the system using a 1200W heat exchanger designed for peak coolant temperatures.

Plant Sciences Inc., Manteca, CA — *Agricultural Robotics Intern* **June 2024 - July 2024**

- Collected field data to evaluate robotic performance in strawberry flower elimination.
- Proposed design modifications to improve mechanical reliability and operational efficiency.
- Digitized commercial farm maps using Microsoft Excel to streamline field logistics.

Weaver's Orchard, Morgantown, PA — *Orchard Maintenance Intern* **June 2023 - Sep 2023**

- Designed and implemented drip irrigation systems for fruit and vegetable fields.
- Operated and maintained agricultural machinery across a 128-acre orchard.
- Performed crop care tasks including pest management, thinning, and harvesting.

Projects:

UC Davis Renewable Grid Design, UC Davis ENG 188 | **Written Report** **March 2025 - June 2025**

- Collaborated to develop a cost-optimized hybrid renewable energy system using NREL's System Advisor Model (SAM) to reduce grid dependency and meet annual energy demand.
- Processed and analyzed hourly annual energy data in Excel to identify the optimal wind-to-solar ratio, reducing supplemental grid electricity use by 70% and LCOE to 7.24¢/kWh.

VHS to Digital Camcorder Conversion, Personal Project | **GrabCAD Model** **Aug 2025 - Sep 2025**

- Designed a custom 12V 2600mAh battery pack and integrated a miniature DVR for analog-to-digital video conversion with a battery life of 6 hours.
- Modified the 10-pin camera connector to accommodate DVR output wires and power distribution.

Vertical Axis Wind Turbine, Personal Project | **GradCAD Model** **Dec 2024 - Jan 2025**

- Designed and assembled a 3D-printed wind turbine prototype using Fusion 360.
- Analyzed performance metrics including wing swept area, rated power, and power curve to evaluate efficiency.

Bike to E-bike Conversion, Personal Project | **GrabCAD Model** **Sep 2024 - Jan 2025**

- Designed and 3D-printed adaptive motor mounts for electric bike conversion with top speeds of 20 mph.
- Applied iterative design process and tested mechanical-electrical integration to achieve a 15 mile range.

Solar Dehydrator, UC Davis EBS 001 | **GrabCAD Model** **Sep 2024 - Dec 2024**

- Collaborated in a team of four to design, build, and test a food dehydration system.
- Analyzed data on moisture loss and presented efficiency findings in a final report.

MATLAB Blackjack App, UC Davis ENG 006 | **GitHub Repository** **Nov 2023 - Dec 2023**

- Developed an interactive blackjack game using MATLAB's GUI framework.
- Met all functionality goals and documented coding process in GitHub.

VEX Battlebot, Governor Mifflin Senior High School, PA **Aug 2021 - May 2022**

- Co-designed and programmed a combat robot using Fusion 360 and C++ for a regional competition.

Skills:

- **Technical Software:** Fusion 360, MATLAB, AutoCAD, SolidWorks, Microsoft Excel, PLC Ladder Logic
- **Data & Analysis:** Data Collection, Statistical Analysis, Technical Writing, Survey Tools
- **Professional:** Team Collaboration, Creative Problem Solving, Adaptability, Communication

Griffin Kriebel - Engineering Portfolio

https://griffinkriebel.com/